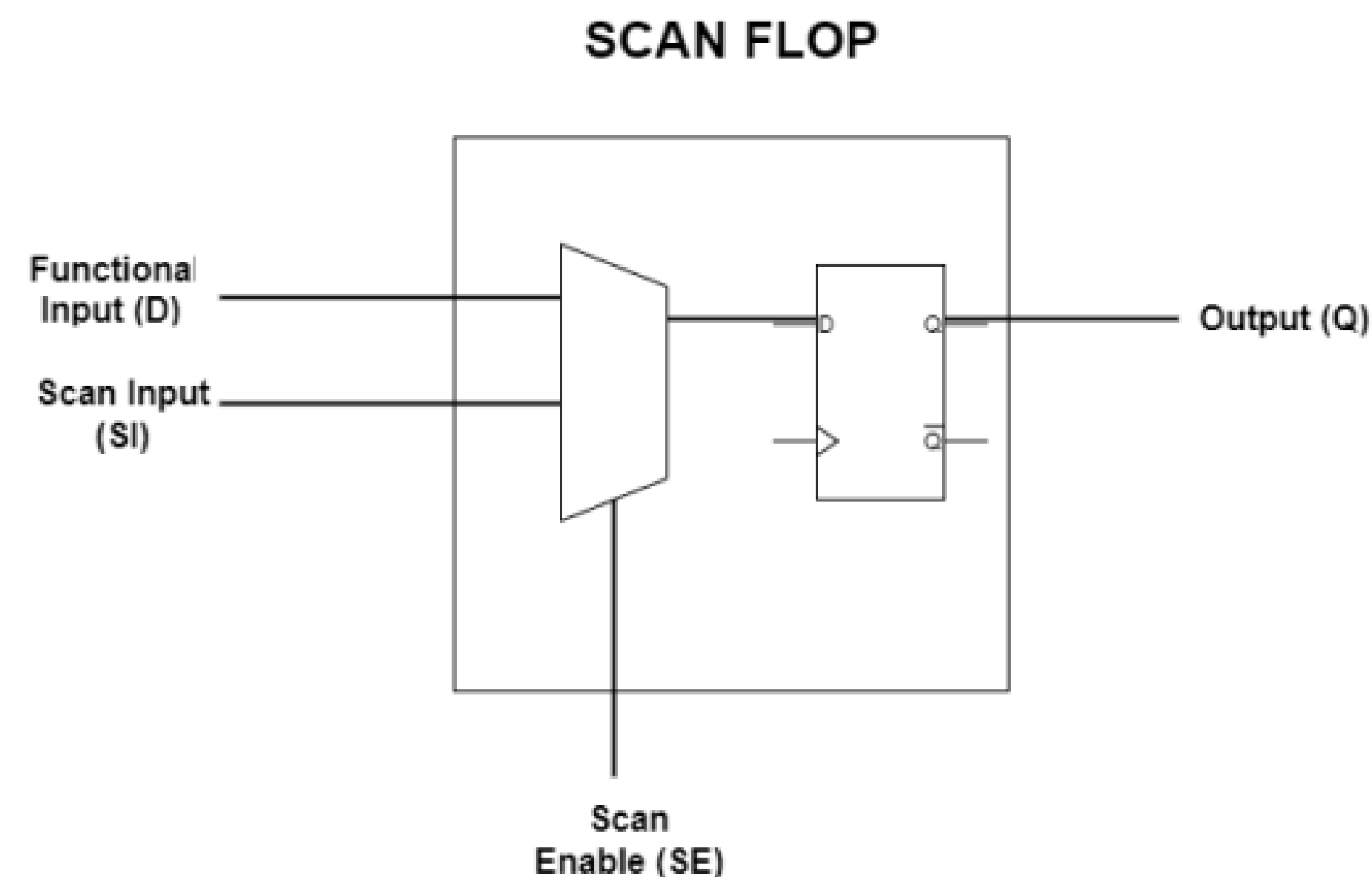


What is Scan Insertion?

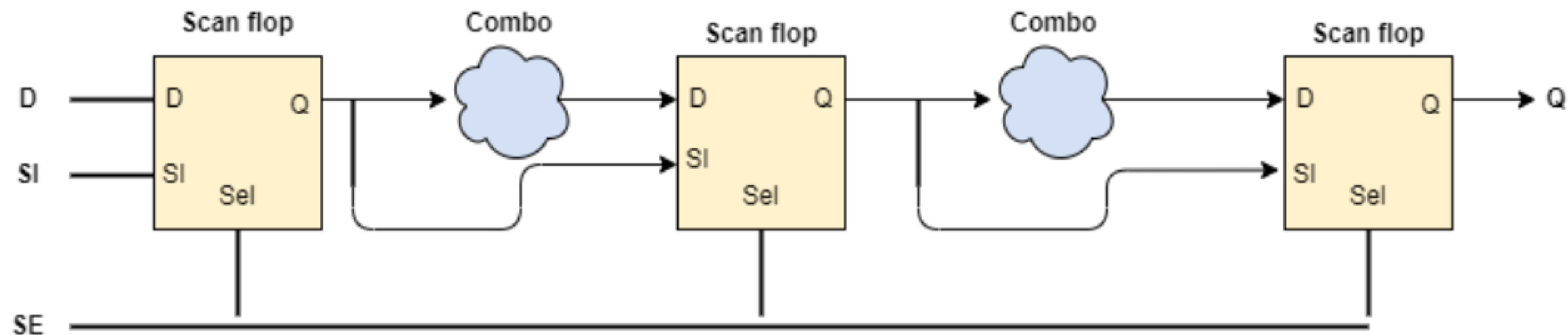
- Scan insertion is the process where all **flops are converted into scan flops** (replaced by their **scan equivalents**) and a shift register path is formed (**q-SI path i.e., scan chain**).



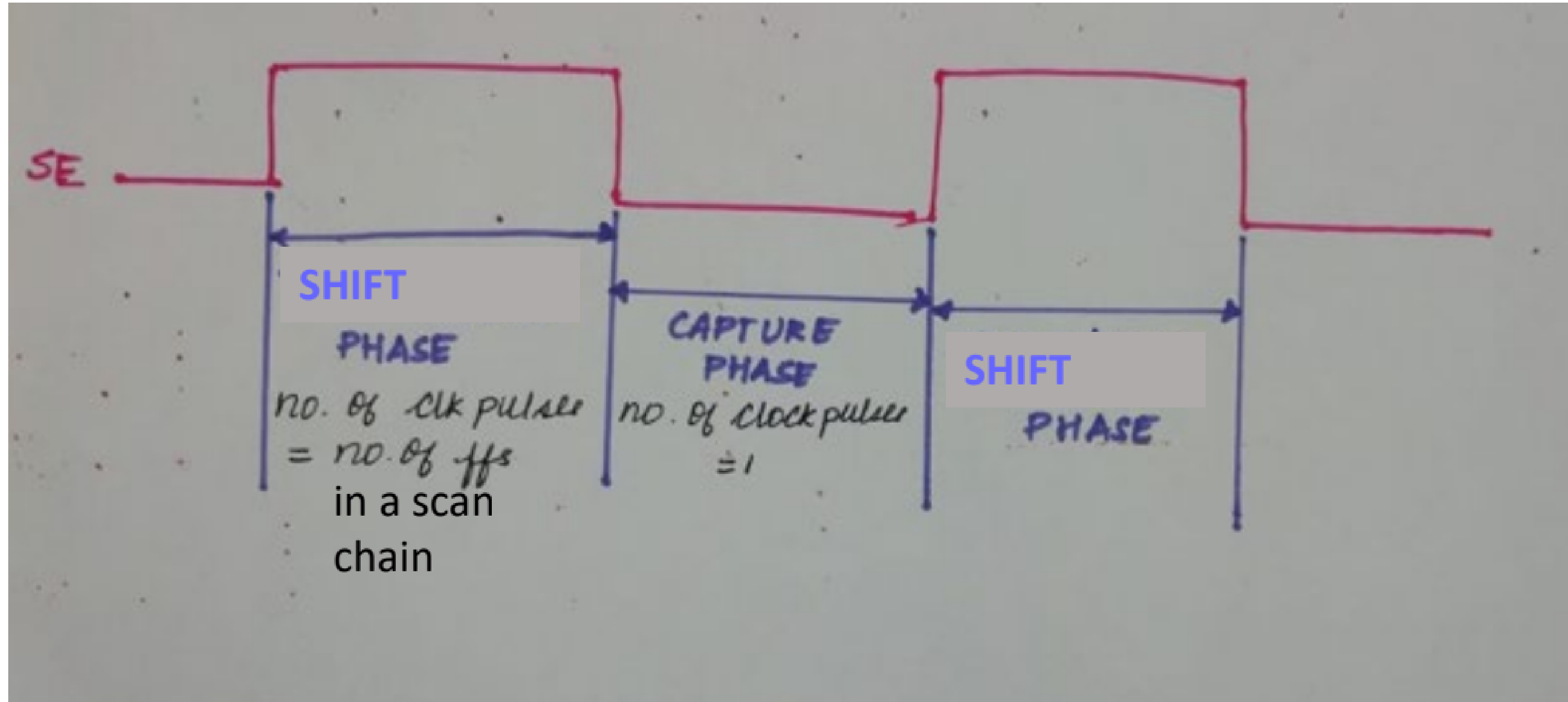
- SE = 0, Functional Input is captured into the flop.
- SE = 1, Scan Input is shifted into the flop.

Scan Chain

SCAN CHAIN



Waveform



- ❑ Scan chain operation involves 3 stages :
 - Scan-in
 - Scan-capture
 - Scan-out
- ❑ Scan-in involves shifting in and loading all the flops with an input vector. During scan-in, the data flows from the output of one flop to the scan-input of the next flop like a shift register.
- ❑ Once the sequence is loaded, one clock pulse (also called as capture pulse) is allowed to excite the combinational logic block and the output is captured at the second flop.
- ❑ The data is then shifted out and the signature is compared with the expected signature.

R. VIDISHA

Need For Scan Insertion

- Scan Insertion is needed because every node in the design should be toggled to gain controllability and observability of the design and this can help in gaining high test coverage of the entire circuit.

Controllability :

The ability to drive the net to '0' and '1' is called controllability

Observability :

The ability to observe the effect of controlling is called observability.

Need For Scan Insertion (Contd...)

EXAMPLE

DESIGN : 10 bit counter

Counter initialization circuit

Counter observation

Suppose if the counter is failing (after counting 999, instead of counting 1000, it again counts 999)

How many clock pulses are required by functional verification team and DFT team to find out that the counter is failing ?

EXAMPLE (Contd...)

Functional Verification

Functional verification team requires more than 1000 pulses to detect that the counter is failing.

Reason : They check functionality

- In this case, counter is failing at the 1000th pulse. So 1000 pulses are required.
- Some clock pulses are also required for counter initialization and counter observation.

DFT

DFT team will be able to tell that the counter is failing approximately in the range of 100 clock pulses.

(Explanation in the next slide)

DFT team will also be able to tell what manufacturing defect has caused the counter to fail.

Need For Scan Insertion (Contd...)

In order to toggle each and every net in the counter to the desired value and to observe it, we need approximately around 6 to 7 patterns.

Suppose we have given :

insert_test_logic –max_length 10 [In 1 scan chain, we can have a maximum of 10 flops]

P1 L – 10 cycles

P1 C – 1 cycle

P1 U, P2 L – 10 cycles

P2 C – 1 cycle

P2 U, P3 L – 10 cycles

P3 C – 1 cycle

P3 U, P4 L – 10cycles

P4 C – 1 cycle

P4 U, P5 L – 10 cycles

P5 C – 1cycle

P5 U, P6 L – 10 cycles

P6 C – 1 cycle

P6 U, P7 L – 10 cycles

P7 C – 1 cycle

P7 U – 10 cycles

Total = 87 cycles

The total no. of cycles can vary slightly based on the scan configuration and no. of patterns required.
So we can tell that counter is failing approximately in the range of 100 clock pulses.

R. VIDISHA

Circuit operation type and scan cell mode

Circuit Operation type	Scan cell mode	TM	SE
Normal	Normal	0	0
Shift Operation	Shift	1	1
Capture Operation	Capture	1	0

- When $TM = 0$, it is functional mode.
- When $TM = 1$, it is DFT mode.
- In DFT mode, SE decides whether it is Shift operation or Capture operation.
 - SE = 1, Shift Operation
 - SE = 0, Capture Operation

- ❖ Shift mode : to shift data into and out of the scan cells.
- ❖ Capture mode : to capture test response into scan cells.

SCAN BENEFITS

The following are benefits of employing a scan strategy:

- **Highly automated process :**

Using scan insertion tools, the process for inserting scan circuitry into a design is highly automated, thus requiring very little manual effort.

- **Highly-effective, predictable method :**

Scan design is a highly-effective, well-understood, and well-accepted method for generating high test coverage for your design.

- **Ease of use:**

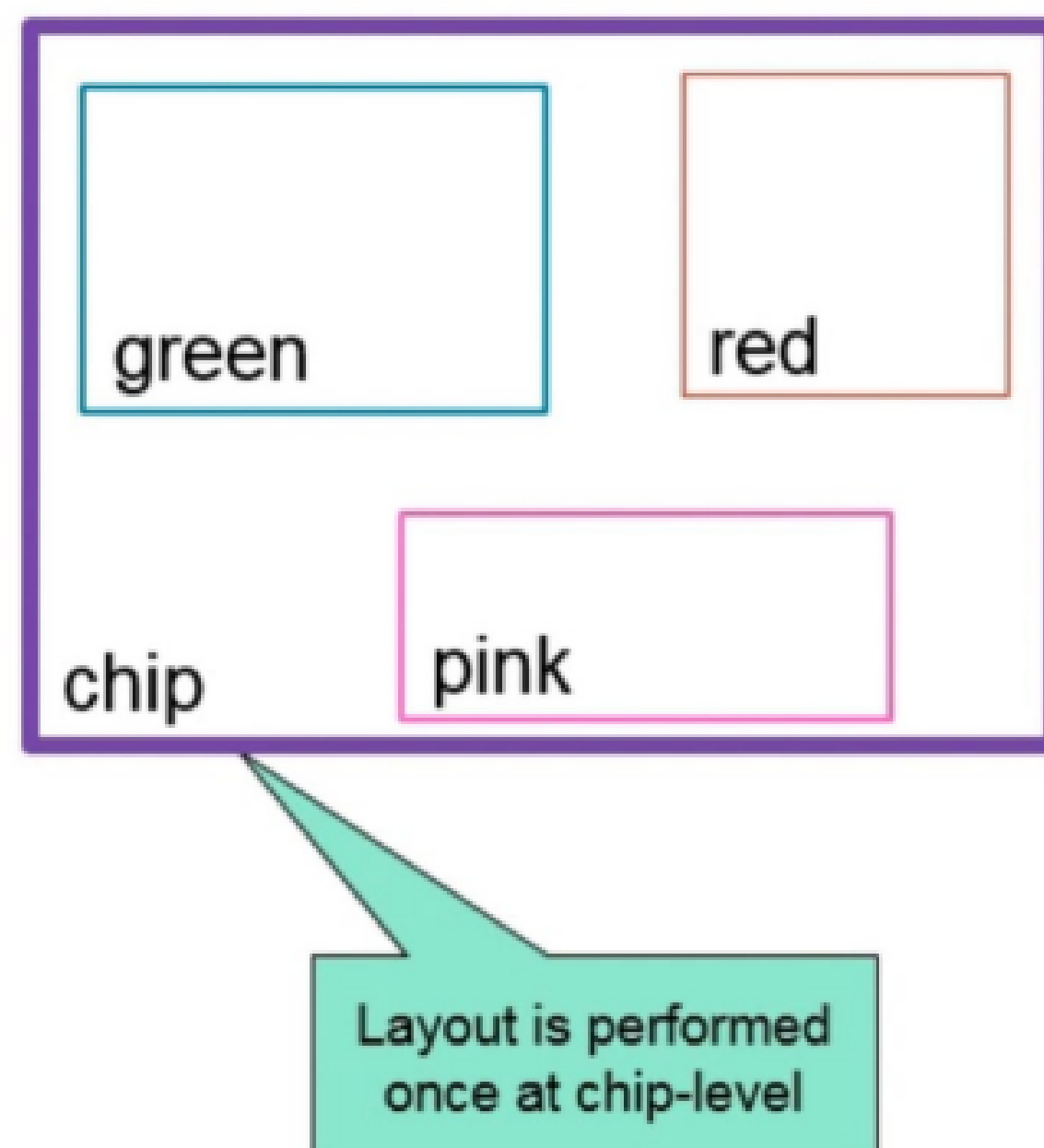
Using scan methodology, you can insert both scan circuitry and run ATPG without the aid of a test engineer.

SCAN METHODOLOGY

- Top-Down (flat) scan methodology
- Bottom-Up (hierarchical) scan methodology

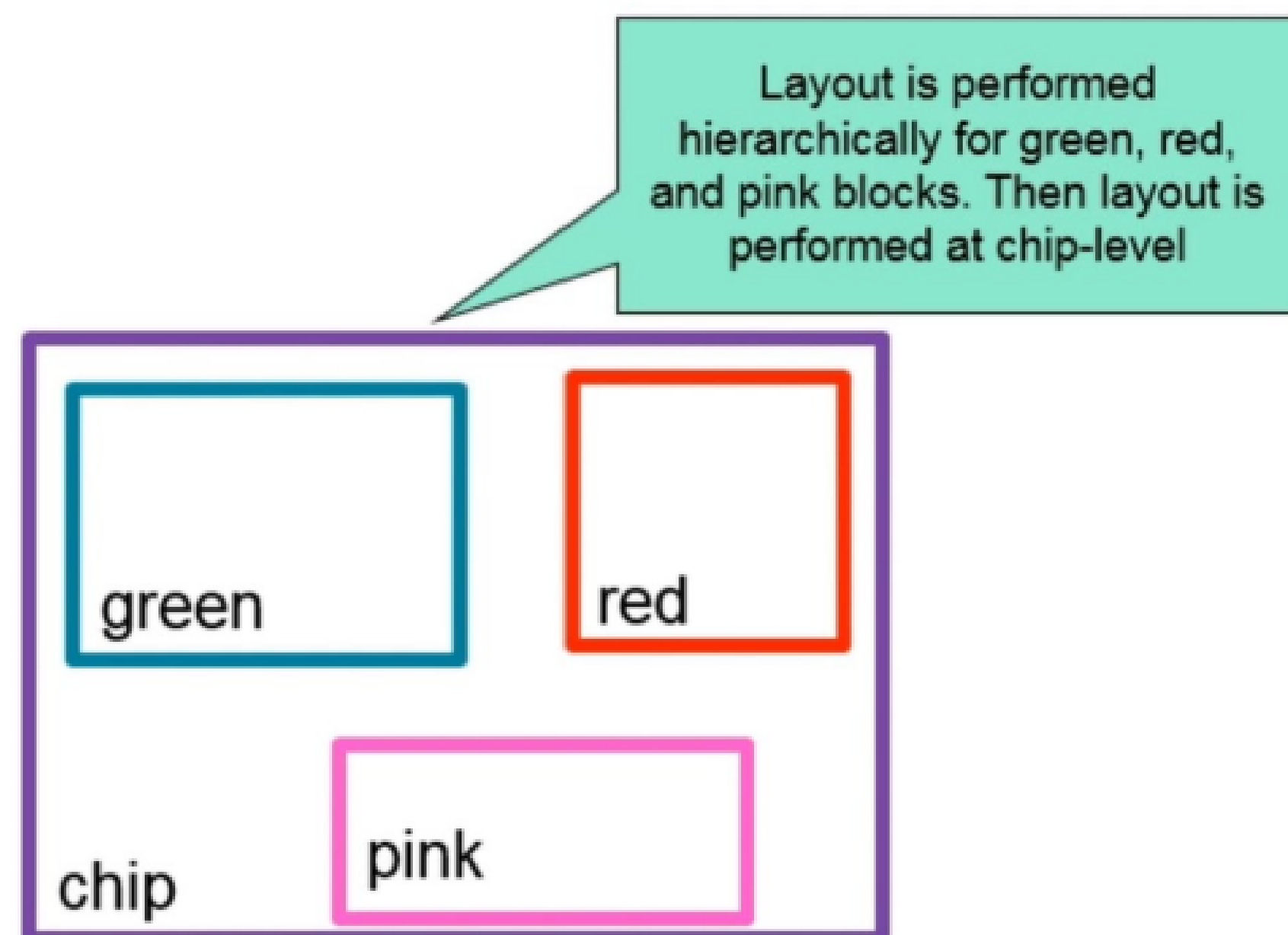
TOP-DOWN (FLAT) METHODOLOGY

- Single DFT insertion operation at the top level of design. This flow is simple, but it requires that DFT insertion be repeated for the entire design if any part of the design changes.
- A very large design that is not suitable for a single top-down DFT insertion
- Time-consuming for large designs



DFT insertion is done once at chip level.

- In bottom-up hierarchical scan synthesis, perform DFT insertion at a lower level of hierarchy, then incorporate those completed scan structures into DFT insertion at a higher level of hierarchy.
- Re-architect DFT structures in a block independently of its surrounding design logic.
- Time-consuming is less for large designs.
- Create a DFT-inserted block that can be reused in future designs.



DFT insertion is performed bottom up for each of the green, red and pink hierarchical layout regions, and then DFT insertion is performed at the chip level.

Full-Scan Design

All storage element are converted into scan cells and combinational ATPG is used for test generation.

Partial-Scan Design

A subset of storage elements are converted into scan cells and sequential ATPG is typically used for test generation

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